

# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

## PROFESSIONAL EXPERIENCE

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### Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained a **Linux Yocto build system**, customizing the kernel and file system for embedded device integration, reducing build times by **18%**.
- Contributed to the development of a **container-based build environment** using Docker, streamlining deployment and improving reproducibility across the team's development lifecycle.
- Engineered a **Python-based AI solution** to automate requirement analysis and code generation for platform deployment, accelerating the initial project setup phase by **30%**.
- Designed and implemented the **Vision framework** for a new TI SoC device, adapting the toolchain and setting up project repositories according to platform standards.
- Developed source code in **C++** adhering to coding guidelines and MISRA standards, and participated in formal code reviews for quality assurance.
- Executed black box, integration, and unit level testing for platform modules, updating software requirement specifications in collaboration with architects and senior developers.
- Managed work products in **CodeBeamer configuration management**, ensuring adherence to company QA requirements and standard software quality processes like ASPICE.

## SKILLS

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**Embedded Software** – firmware, drivers, middleware, low level software

**Programming Languages** – C, C++, Python

**Hardware-Software Interface** – GPIOs, SPI, I2C, UART, Ethernet, Memory

**Build & Toolchain** – Linux Yocto, CMake, Compiler, Linker

**Documentation** – Wiki, Google Sites

**Development Process** – ASPICE, ISO15504, MISRA, static code analysis, Klocwork, CodeBeamer

**Containerization** – Docker

## PROJECTS

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### Embedded Linux Platform for Automotive SoC

- Configured and customized the **Linux Yocto build environment** for a new Texas Instruments SoC device, integrating custom kernel modules and file systems.
- Developed middleware components in **C++** to interface with hardware peripherals such as SPI and I2C, ensuring reliable data acquisition.
- Documented the build process and platform setup using **Wiki pages**, facilitating knowledge transfer for new team members joining the project.

### Automated AI Code Generation for Embedded Systems

- Built a **Python-based AI pipeline** that analyzes software requirement specifications and generates initial C++ source code skeletons, reducing manual coding effort by an estimated **25%**.
- Integrated the AI code generation tool into the existing CI/CD workflow using **Docker containers**, enabling rapid iteration and testing of generated code.
- Developed a proof-of-concept demonstrating the AI solution on a target hardware platform, showcasing its utility for platform deployment into customer projects.

## EDUCATION

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### M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

### B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

## CERTIFICATIONS

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- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)